

Independent external-marker neighborhood preservation

Fixed external marker panels; all eligible annotation strata; marker selection did not use an evaluated embedding.

Dataset	Backbone	Representation	AUROC	Average precision (AP)	Contrast
HP-CB	Scanorama	Upstream	0.9873	0.9349	1.5068
		GenoRefine	0.9824 [0.9782, 0.9851]	0.8770 [0.8606, 0.8904]	1.4111 [1.3744, 1.4398]
		IDEC	0.9432 [0.9276, 0.9578]	0.7301 [0.6866, 0.8025]	1.0925 [1.0143, 1.2348]
HP-CB	Harmony	Upstream	0.9863	0.9225	1.4180
		GenoRefine	0.9852 [0.9843, 0.9862]	0.9271 [0.9239, 0.9303]	1.4148 [1.4120, 1.4173]
		IDEC	0.9802 [0.9792, 0.9814]	0.8917 [0.8873, 0.8966]	1.2997 [1.2948, 1.3115]
HP-CB	Seurat	Upstream	0.9872	0.9312	1.4701
		GenoRefine	0.9865 [0.9857, 0.9871]	0.9230 [0.9205, 0.9261]	1.4748 [1.4692, 1.4814]
		IDEC	0.9856 [0.9844, 0.9872]	0.9207 [0.9131, 0.9261]	1.4065 [1.3853, 1.4186]
HP-CB	Online iNMF	Upstream	0.9750	0.8436	1.3540
		GenoRefine	0.9744 [0.9737, 0.9753]	0.8450 [0.8430, 0.8471]	1.3503 [1.3438, 1.3544]
		IDEC	0.9677 [0.9596, 0.9738]	0.8348 [0.8309, 0.8409]	1.2779 [1.2604, 1.2912]
Mouse	Scanorama	Upstream	0.8498	0.1607	0.8389
		GenoRefine	0.8594 [0.8555, 0.8645]	0.1381 [0.1324, 0.1477]	0.6788 [0.6057, 0.7294]
		IDEC	0.8530 [0.8453, 0.8605]	0.1250 [0.1139, 0.1350]	0.5785 [0.5260, 0.6175]
Mouse	Harmony	Upstream	0.8391	0.1492	0.7236
		GenoRefine	0.8404 [0.8397, 0.8412]	0.1453 [0.1443, 0.1465]	0.7245 [0.7176, 0.7312]
		IDEC	0.8423 [0.8406, 0.8443]	0.1413 [0.1396, 0.1458]	0.6670 [0.6627, 0.6712]
Mouse	Seurat	Upstream	0.8408	0.1506	0.7689
		GenoRefine	0.8429 [0.8414, 0.8440]	0.1493 [0.1469, 0.1513]	0.7710 [0.7663, 0.7730]
		IDEC	0.8406 [0.8393, 0.8418]	0.1433 [0.1413, 0.1446]	0.7060 [0.6848, 0.7182]
Mouse	Online iNMF	Upstream	0.8613	0.1494	0.6809
		GenoRefine	0.8627 [0.8624, 0.8630]	0.1491 [0.1475, 0.1502]	0.6975 [0.6839, 0.7153]
		IDEC	0.8597 [0.8586, 0.8615]	0.1281 [0.1230, 0.1332]	0.5934 [0.5724, 0.6313]

Upstream is a fixed embedding. GenoRefine and IDEC are mean [minimum, maximum] across five algorithmic seeds; seeds are not biological replicates.